

- 6 (a) The bottlenose dolphin, *Tursiops truncatus*, is an aquatic mammal. It has adaptations to conserve heat when swimming in cold water.

Fig. 6.1 shows a bottlenose dolphin.

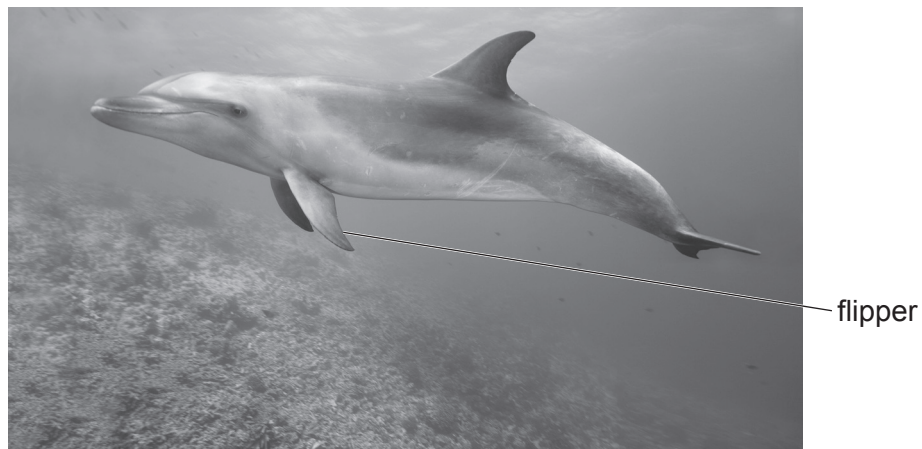


Fig. 6.1

- (i) The arteries in the bottlenose dolphin that carry blood to the flippers are surrounded by veins bringing blood back to the rest of the body.

Outline how this arrangement of blood vessels is an adaptation to conserve body heat.

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..... [1]

- (ii) When a dolphin has exercised for a long time, it may need to lose excess heat. Unlike humans, dolphins do **not** produce sweat.

Suggest how a dolphin may lose excess heat.

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..... [1]

(b) The control of blood glucose concentration in dolphins is the same as in most mammals.

A study was carried out to investigate the concentration of insulin in the blood of dolphins that were provided with a diet of fish, supplemented with glucose.

- One group of dolphins ate fish with 3 g of glucose for every kg of fish.
- Another group ate fish with 11 g of glucose for every kg of fish.

The concentration of insulin in the blood was measured 60 minutes before being fed, at the time of being fed and at regular intervals afterwards.

Measurements were also taken for a third group of dolphins, at the same time intervals, that were not fed any fish (fasting).

The results of the study are shown in Fig. 6.2.

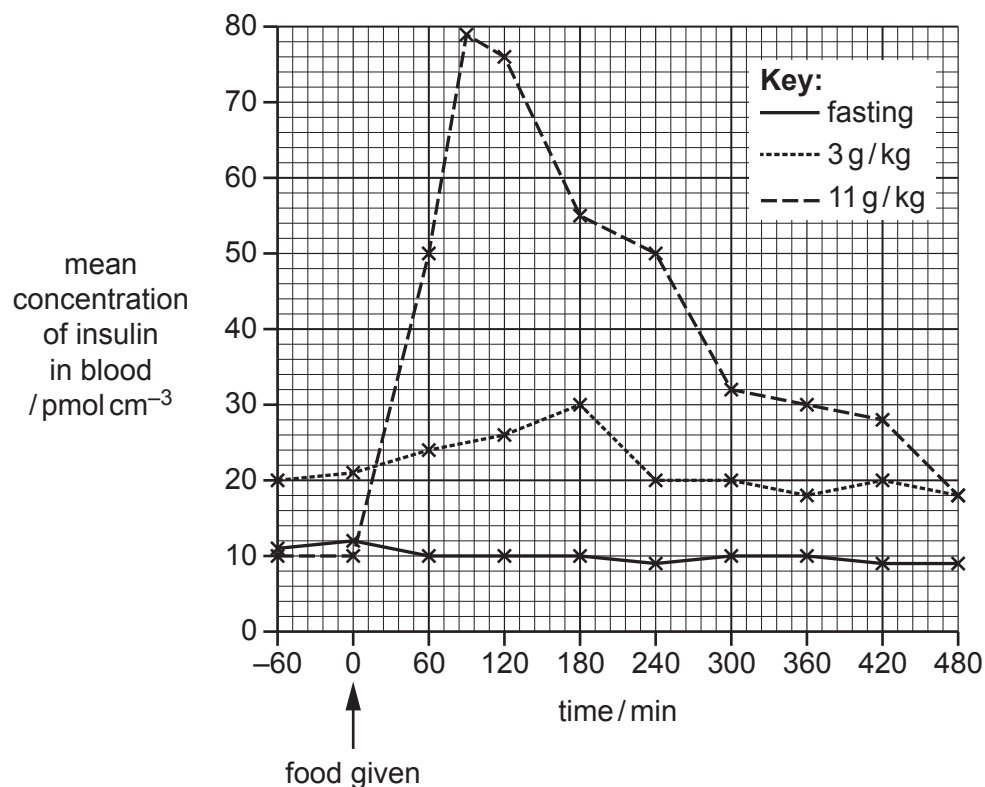


Fig. 6.2

(i) Describe the trends shown in Fig. 6.2.

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(ii) The mean concentration of insulin in the blood changes over time for the dolphins that were fed fish with 11 g of glucose per kg of fish.

Calculate the rate of change in the mean concentration of insulin in this group of dolphins from the time of being fed until the concentration reaches its maximum.

Show your working.

..... $\text{pmol cm}^{-3} \text{min}^{-1}$ [2]

(c) Blood glucose concentration is regulated by negative feedback.

Explain what is meant by negative feedback.

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..... [3]